本征函数：
- Helmholtz方程：
  - 柱：$J_\nu(x) \to J_n(k_{ni}r)$
  - 球：$j_\nu(x) \to j_l(k_{nl}r)$
- $\nabla^2 u$ 在球坐标下
  - 径向同性：$P_l(x) \to P_l(\cos\theta)$
  - $P_l^m(x) \to P_l^m(\cos\theta)$，$m=-l,-l+1,\cdots,l-1,l$，$l=0,1,2,\cdots$

要记表达：$P_l(x)=\frac{1}{2^l l!}\frac{d}{dx^l}((x^2-1)^l)$ → $P_l(x)$ 为 x 的 l 次多项式，最高次系数为 $\frac{(2l)!}{2^l l!}$　$P_{2l}(0)=(-1)^l\frac{(2l)!}{2^l(l!)^2}$，$P_{2l+1}(0)=0$

$P_0(x)=1$，$P_1(x)=x$，$P_2(x)=\frac{1}{2}(3x^2-1)$，$P_l(1)=1$

$P_l^m(x)=(-1)^m(1-x^2)^{\frac{m}{2}}P_l^{(m)}(x)$ → $P_l^{-m}(x)=(-1)^m\frac{(l-m)!}{(l+m)!}P_l^m(x)$

$J_\nu(x)=\sum_{k=0}^{\infty}\frac{(-1)^k}{k!\Gamma(k+\nu+1)}\left(\frac{x}{2}\right)^{2k+\nu}$ → $J_{-l}(x)=(-1)^l J_l(x)$

$j_l(x)=\sqrt{\frac{\pi}{2x}}J_{l+\frac{1}{2}}(x)$，$j_0(x)=\frac{\sin x}{x}$　　$J_0'=-J_1$

递推关系：$P_l(x)$：
$$\begin{cases}\frac{d}{dx}(x^{\nu+1}J_{\nu+1}(x))=x^{\nu+1}J_\nu(x)\\ \frac{d}{dx}(x^{-\nu}J_\nu(x))=-x^{-\nu}J_{\nu+1}(x)\end{cases}\Rightarrow\begin{cases}J_{\nu-1}-J_{\nu+1}=2J_\nu'\\ J_{\nu-1}+J_{\nu+1}=\frac{2\nu}{x}J_\nu\end{cases}\qquad\begin{cases}P_{l+1}'(x)=xP_{l+1}'(x)+(l+1)P_l'\\ P_{l-1}'(x)=xP_l'-lP_l(x)\end{cases}$$

生成函数：$\frac{1}{\sqrt{1-2xt+t^2}}=\sum_{l=0}^{\infty}P_l(x)t^l$，$e^{\frac{z}{2}(t-\frac{1}{t})}=\sum_{n=0}^{\infty}J_n(z)t^n$

引理：$\int_x^1 P_k(x)P_l(x)\,dx=\frac{(1-x^2)(P_k'P_l-P_kP_l')}{k(k+1)-l(l+1)}$，$\int_0^x\frac{J_l(x)J_m(x)}{x}dx=\frac{x(J_m'J_m-J_mJ_m')}{m^2-n^2}$

$J_n(x+y)=\sum_{k=-\infty}^{\infty}J_k(x)J_{n-k}(y)$

正交归一性：$\int_{-1}^1 P_k(x)P_l(x)dx=\frac{2}{2l+1}\delta_{kl}\Rightarrow\int_{-1}^1 P_k(\cos\theta)P_l(\cos\theta)\sin\theta\,d\theta=\frac{2}{2l+1}\delta_{kl}$

$\int_{-1}^1 P_k^m(x)P_l^m(x)dx=\frac{2}{2l+1}\frac{(l+m)!}{(l-m)!}\delta_{kl}$，$\int_{-1}^1 P_l^m(x)P_k^{-m}(x)dx=\frac{(-1)^m\cdot 2}{2l+1}\delta_{kl}$

$\int_0^a J_n(k_{ni}r)J_n(k_{nj}r)\cdot r\,dr=\frac{a^2}{2}(J_{n+1}(\mu_n^i))^2\delta_{ij}$

$\int_0^a j_l(k_{ln}r)j_l(k_{ln}r)r^2dr=\frac{a^3}{2}(j_l'(\mu_l^i))^2\delta_{ij}$

$\int_0^b\sin\frac{n\pi}{l}x\sin\frac{m\pi}{l}x\,dx=\frac{b}{2}\delta_{nm}$，$\int_0^{2\pi}\sin m\varphi\sin n\varphi\,d\varphi=\pi\delta_{nm}$

奇偶性：$P_l(-x)=(-1)^lP_l(x)$，$J_{-n}(x)=(-1)^nJ_n(x)$

本征方程的一般解：Laplace方程：$\nabla^2 u=0$

极坐标下：$\frac{1}{r}\frac{\partial}{\partial r}(r\frac{\partial u}{\partial r})+\frac{1}{r^2}\frac{\partial^2 u}{\partial \varphi^2}=0$

$$\Rightarrow \begin{cases} \Phi''+\lambda\Phi=0 \\ \frac{1}{r}\frac{d}{dr}(r\frac{dR}{dr})-\frac{\lambda R}{r^2}=0 \end{cases} \Rightarrow \begin{cases} \Phi=e^{im\varphi},\ \lambda=m^2,\ m=1,2,\cdots \\ R=A_m r^m+B_m r^{-m} \end{cases}$$

$m=0$时：$u=C_0+D_0\ln r+\sum_{m=1}^{\infty}\left[(C_m r^m+D_m r^{-m})\sin m\varphi+(C_m r^m+D_m r^{-m})\cos m\varphi\right]$，周期：$D_m=0$

球坐标下：$\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial u}{\partial r})+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\frac{\partial u}{\partial\theta})+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 u}{\partial\varphi^2}=0 \Rightarrow \frac{1}{r}\frac{d}{dr}(r\frac{dR}{dr})-\frac{m^2}{r^2}R=f$

$$\Rightarrow \begin{cases} \frac{1}{r}\frac{d}{dr}(r^2\frac{dR}{dr})-\frac{1}{r^2}\lambda R=0 \\ \Phi''+\mu\Phi=0 \\ \sin\theta\frac{d}{d\theta}(\sin\theta\frac{d\Theta}{d\theta})+(\lambda-\frac{\mu}{\sin^2\theta})\Theta=0 \end{cases} \Rightarrow \begin{cases} \Phi=e^{im\varphi},\ \mu=m^2,\ m=0,1,2,\cdots \\ R=A_l r^l+B_l r^{-l-1} \\ \Theta=P_l^m(\cos\theta),\ \lambda=l(l+1),\ l=0,1,2,\cdots \end{cases}$$

$\Rightarrow$一般解：$u=\sum_{m=0}^{\infty}\sum_{l=0}^{\infty}(A_l r^l+B_l r^{-l-1})P_l^m(\cos\theta)e^{im\varphi}$，球土或：$B_l=0 \Rightarrow \frac{1}{r^2}\frac{d}{dr}(r^2\frac{dR}{dr})-\frac{l(l+1)}{r^2}R=f$

Helmholtz方程：$\nabla^2 u+k^2 u=0$

极坐标：$\frac{1}{r}\frac{\partial}{\partial r}(r\frac{\partial u}{\partial r})+\frac{1}{r^2}\frac{\partial^2 u}{\partial\varphi^2}+k^2 u=0$

$$\Rightarrow \begin{cases} \Phi''+\lambda\Phi=0 \\ \frac{1}{r}\frac{d}{dr}(r\frac{dR}{dr})+(k^2-\frac{\lambda}{r^2})R=0 \end{cases} \Rightarrow \begin{cases} \Phi=e^{im\varphi},\ \lambda=m^2,\ m=0,1,2,\cdots \\ R=J_m(\frac{\mu_i^{(m)}}{a}r),\ k_{mi}=\frac{\mu_i^{(m)}}{a},\ i=1,2,3,\cdots \end{cases}$$

一般解：$u=\sum_{m=0}^{\infty}\sum_{i=1}^{\infty}C_{im}J_m(\frac{\mu_i^{(m)}}{a}r)e^{im\varphi}$

球坐标：$\frac{1}{r^2}\frac{\partial}{\partial r}(r^2\frac{\partial u}{\partial r})+\frac{1}{r^2\sin\theta}\frac{\partial u}{\partial\theta^2}+\frac{1}{r^2\sin^2\theta}\frac{\partial^2 u}{\partial\varphi^2}+k^2 u=0$

$$\Rightarrow \begin{cases} \Phi''+\mu\Phi=0 \\ \frac{1}{r^2}\frac{d}{dr}(r^2\frac{dR}{dr})+(k^2-\frac{\lambda}{r^2})R=0 \\ \sin\theta\frac{d}{d\theta}(\sin\theta\frac{d\Theta}{d\theta})+(\lambda-\frac{\mu}{\sin^2\theta})\Theta=0 \end{cases} \Rightarrow \begin{cases} \Phi=e^{im\varphi},\ \mu=m^2,\ m=-l,-l+1,\cdots l-1,l \\ R=j_l(\frac{\mu_i^{(l)}}{a}r),\ k_{ni}=\frac{\mu_i^{(l)}}{a},\ i=1,2,3,\cdots \\ \Theta=P_l^m(\cos\theta),\ \lambda=l(l+1),\ l=0,1,2,\cdots \end{cases}$$

一般解：$u(r,\theta,\varphi)=\sum_{i=1}^{\infty}\sum_{l=0}^{\infty}\sum_{m=-l}^{l}(C_{ilm}j_l(\frac{\mu_i^{(l)}}{a}r)P_l^m(\cos\theta)e^{im\varphi}$

含时齐次的一般解：（默认边界齐次）

波动方程：$\frac{\partial^2 u}{\partial t^2} - c^2 \nabla^2 u = 0$

令 $u = V(r,\theta,\varphi) e^{i\omega t} \Rightarrow$ Helmholtz 方程：$\nabla^2 u + k^2 u = 0$，其中 $k = \frac{\omega}{c} \to e^{ik^2 t}$

一般解：极坐标：$u(r,\varphi,t) = \sum_{m=0}^{\infty} \sum_{i=1}^{\infty} C_{mi} J_m(\frac{\mu_i^{(m)}}{a} r) e^{im\phi} e^{i\frac{\mu_i^{(m)}}{a} ct}$

球坐标：$u(r,\theta,\varphi,t) = \sum_{l=0}^{\infty} \sum_{i=1}^{\infty} \sum_{m=-l}^{l} C_{ilm} j_l(\frac{\mu_i^{(l)}}{a} r) P_l^m(\cos\theta) e^{im\phi} e^{i\frac{\mu_i^{(l)}}{a} ct}$

热传导方程：$\frac{\partial u}{\partial t} - k \nabla^2 u = 0$

令 $u = V e^{-\omega t} \Rightarrow$ Helmholtz 方程 $\nabla^2 u + k^2 u = 0$，其中 $k^2 = \frac{\omega}{k} \to e^{-k k^2 t}$

一般解：极坐标：$u(r,\varphi,t) = \sum_{m=0}^{\infty} \sum_{i=1}^{\infty} C_{mi} J_m(\frac{\mu_i^{(m)}}{a} r) e^{im\phi} e^{-k(\frac{\mu_i^{(m)}}{a})^2 t}$

球坐标：$u(r,\theta,\varphi,t) = \sum_{l=0}^{\infty} \sum_{i=1}^{\infty} \sum_{m=-l}^{l} C_{ilm} j_l(\frac{\mu_i^{(l)}}{a} r) P_l^m(\cos\theta) e^{im\phi} e^{-k(\frac{\mu_i^{(l)}}{a})^2 t}$

若 $u|_{t=0} = u_0 \Rightarrow u_0 = \sum_{i=1}^{\infty} C_i j_0(\frac{\mu_i^{(0)}}{a} r) = \sum_{i=1}^{\infty} C_i \frac{\sin\frac{i\pi}{a} r}{\frac{i\pi}{a} r} \Rightarrow C_i = 2u_0(-1)^{n+1}$

$$\Rightarrow u = \sum_{i=1}^{\infty} 2u_0(-1)^{n+1} j_0(\frac{i\pi}{a} r) e^{-k\frac{i^2\pi^2}{a^2} t}$$

柱坐标：Laplace 方程 $\frac{1}{r}\frac{d}{dr}(r\frac{du}{dr}) + \frac{1}{r^2}\frac{d^2u}{d\varphi^2} + \frac{d^2u}{dz^2} = 0$

$$\Rightarrow \begin{cases} \Phi''(\varphi) + \lambda \Phi(\varphi) = 0 \\ Z'' + \lambda Z = 0 \\ \frac{1}{r}\frac{d}{dr}(r\frac{dR}{dr}) + (-\lambda - \frac{\nu^2}{r^2})R = 0 \end{cases} \Rightarrow \begin{cases} \Phi = e^{im\phi}, m = 0, 1, 2, \cdots \\ Z = \sin\frac{n\pi}{h} z, n = 1, 2, 3, \cdots \\ R = I_m(\frac{n\pi}{h} r) \end{cases}$$

一般解：$u = \sum_{m=0}^{\infty} \sum_{n=1}^{\infty} C_{mn} I_m(\frac{n\pi}{h} r) e^{im\phi} \sin\frac{n\pi}{h} z$

本征值问题：

Bessel函数：生成函数、奇偶性、递推关系、引理

$$e^{\frac{x}{2}(t-\frac{1}{t})}=\sum_{n=-\infty}^{\infty}J_n(x)t^n \Rightarrow e^{ix\sin\theta}=\sum_{n=-\infty}^{\infty}J_n(x)e^{in\theta}=J_0+2\sum_{n=1}^{\infty}J_{n(x)}i^n\cos n\theta$$

$$J_n(x)=\frac{1}{2\pi}\int_{-\pi}^{\pi}e^{i(x\sin\theta-n\theta)}d\theta$$

$$J_n(x+y)=\sum_{k=0}^{\infty}J_k(x)J_{n-k}(y)$$

$$\int_{-1}^{1}x^{l+1}P_l(x)dx=2^{l+1}\frac{(l+2n)!(l+n)!}{n!(2l+2n+1)!}$$

S-L问题：

$$\frac{d}{dx}(p(x)\frac{dy}{dx})+[\lambda\rho(x)-q(x)]y=0 \Rightarrow \text{边界}: p(x)(y_1^*y_2'-y_2y_1^{*\prime})\Big|_a^b=0$$

$$\text{正交}: \int_{-1}^{1}y_1(x)y_2(x)\rho(x)dx=0$$

一维Fourier变换：

$$F(k)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}f(x)e^{-ikx}dx,\quad f(x)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}F(k)e^{ikx}dk$$

$$f^{(n)}(x)=(ik)^nF(k),\quad F(k)G(k)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}f(\varphi)g(x-\varphi)d\varphi$$

含时Green函数：

$U: t\to t' \Rightarrow G: t\to -t'$；

$t'=y(x')\Rightarrow J_{t',t}=0$